

KAREN JEFFREY

Political Economist | Data Scientist

karenjeffrey.github.io

SUMMARY

I am a Political Economist and Data Scientist with expertise in survey design, causal inference, and predictive modelling. I have completed a PhD in Political Economy, a Postdoc in Medical Informatics, and have a track record of delivering impactful, policy-relevant research.

EMPLOYMENT

2026 - Data Science & AI Fellow, University of Glasgow

Leading a program of research on the impacts of AI on labour markets and policy responses.

2022 - 2024 Research Fellow in Medical Informatics, University of Edinburgh

Conducted award-winning Medical Informatics research. Applied advanced data science techniques to analyse world class health data. Collaborated with clinicians, patients, and policy-makers.

2021 - 2022 Senior Intelligence Analyst in Public Health, NHS Lothian

Led the development of the Lothian Public Health Survey. Gathered, analysed, and interpreted data and communicated findings to support Public Health planning, policy design, and service delivery.

2018 - 2020 Teaching Assistant in Political Economy, King's College London

Led seminars, held office hours, and marked assignments for undergraduate students studying The Political Economy of Inequality and Microeconomics 1.

2017 - 2018 Senior Research Consultant, BBC Media Action

Managed evaluations of BBC Media Action's international development interventions. Funders included: UNICEF, FCO, DFID, Global Affairs Canada.

2013 - 2016 Researcher, New Economics Foundation (Think tank)

Developed and lobbied for the adoption of new measures of national success. Initiated a new research programme on "The Future of Work." Funders included: European Commission, Co-op, Kingfisher.

2011 - 2013 Analyst, Delta Energy and Environment (Consultancy)

Conducted research and modelling to forecast the impact of policy and technological developments on markets for renewable energy technologies. Clients included: DECC, E.On, British Gas, Bosch.

EDUCATION

2017 - 2021 PhD Political Economy (Pass with no corrections), King's College London

Thesis: *Essays on the Political Economy of Automation*.

2009 - 2010 M.Sc. Ecological Economics (Distinction), University of Edinburgh

2003 - 2007 B.A. (Hons.) Media (2:1), Queen Margaret University

POLITICAL ECONOMY PUBLICATIONS

Jeffrey, K. (2024). [Measuring Automation’s Impact on Labor Markets: A new approach is needed.](#)

Jeffrey, K. & Matakos, K. (2024). [Automation anxiety, fairness perceptions, and redistribution: How past experience conditions the response to future job loss.](#) *Journal of Economic Behavior & Organization*.

Jeffrey, K. (2021). [Automation and the future of work: how rhetoric shapes the response in policy preferences.](#) *Journal of Economic Behavior & Organization*.

Jeffrey, K. (2021). [Measuring exposure to automation: a survey approach.](#)

MEDICAL INFORMATICS PUBLICATIONS

Shah, S. A., **Jeffrey, K.**, Robertson, C., Sheikh, A. (2025). [Impact of COVID-19 pandemic on elective care backlog trends, recovery efforts, and capacity needs to address backlogs in Scotland \(2013–2023\).](#) *The Lancet Regional Health Europe*.

McCabe, R., Kibuchi, E., Amele, S., et al. (2025). [Effect modification and interaction between ethnicity and socioeconomic factors in severe COVID-19.](#) *BMJ Open*.

Jeffrey, K., Hammersley, V., Maini, R., et al. (2024). [Deriving and validating a risk prediction model for long COVID: population-based, retrospective cohort study in Scotland.](#) *Journal of the Royal Society of Medicine*.

Jeffrey, K., Woolford, L., Maini, R., et al. (2024). [Identifying long COVID using electronic health records: a national observational cohort study in Scotland.](#) *eClinicalMedicine*.

Shi, T., Millington, T., Robertson, C., et al. (2024). [Risk of winter hospitalisation and death from acute respiratory infections in Scotland.](#) *Journal of the Royal Society of Medicine*.

Millington, T., Morrison, K., **Jeffrey, K.** et al. (2024). [Caveats in reporting of national vaccine uptake.](#) *Journal of Global Health*.

Amele, S., McCabe, R., Kibuchi, E., et al. (2024). [Quality of ethnicity data within Scottish health records.](#) *Journal of Public Health*.

Macdonald, C., Palmateer, N., McAuley, A., et al. (2024) [Association between antibody response post-vaccination and severe COVID-19 outcomes.](#) *npj Vaccines*.

Kerr, S., Greenland, S., **Jeffrey, K.**, et al. (2023). [Understanding and reporting odds ratios and rate ratios in case-control studies.](#) *Journal of Global Health*.

Morrison, K., Cullen, L., James, A., et al. (2023). [Predictors of incomplete COVID-19 vaccine schedule among adults in Scotland](#) *Vaccine*

Kerr, S., Millington, T., Rudan, I., et al. (2023). [External validation of the QCovid 2 and 3 risk prediction algorithms for risk of COVID-19 hospitalisation and mortality in adults.](#) *BMJ Open*.

SELECTED THINK TANK REPORTS

Jeffrey, K., Abdallah, S. & Michaelson, J. (2017). [The Cost of Loneliness to UK Employers](#). London: New Economics Foundation.

Jeffrey, K., Wheatley, H. & Abdallah, S. (2016). [The Happy Planet Index 2016: A Global Index of Sustainable Wellbeing](#). London: New Economics Foundation.

Jeffrey, K. and Michaelson, J. (2015). [Five Headline Indicators of National Success: A Clearer Picture of How the UK is Performing](#). London: New Economics Foundation.

Jeffrey, K., Mahony, S., Michaelson, J. and Abdallah, S. (2014). [Well-being at Work: A Review of the Literature](#). London: New Economics Foundation.

ACHIEVEMENTS & AWARDS

- 2023 **Royal Statistical Society's Florence Nightingale Award** for excellence in healthcare data analytics.
- 2023 **Royal Society of Edinburgh's Mary Somerville Medal** awarded to the EAVE II Team for exceptional achievement in teamwork and collaborative endeavour.
- 2013-24 **Press coverage** of research on long COVID ([online](#), TV, and radio), the future of work ([online](#)), and reports produced while working at the New Economics Foundation (BBC Daily Mail, Financial Times, Herald, Independent, ITV, Observer, Telegraph, Forbes and press in more than 30 countries worldwide).

GRANTS, FELLOWSHIPS & FUNDRAISING

- 2026 **Small Research Grant** BA/Leverhulme (£10,000)
- 2024 **Usher Institute Small Grant** University of Edinburgh (£650)
- 2022 **Data Skills for Work** The Data Lab (£300)
- 2020 **London Interdisciplinary Social Sciences Scholarship** ESRC (£9,000)
Small Academic Expenses Grant Royal Economic Society (£450)
Post-Graduate Research Bursary Grants King's College London (£1,000)
- 2019 **Post-Graduate Research Bursary Grants** King's College London (£500)
Cohort Activity Grant ESRC (£1,600)
- 2018 **Research Development Fund** King's College London (£2,000)
- 2017 **London Interdisciplinary Social Sciences Doctoral Scholarship** ESRC (£68,000)
Bentham PhD Fellowship University College London (£64,000, declined)
- 2011-16 Contributed to successful tenders for funding from European Commission, E.On, British Gas, Vattenfall, Bosch, Vaillant, DECC, Co-op, & Kingfisher Group.

TEACHING & SUPERVISION

- 2018-20 **Teaching Assistant** The Political Economy of Inequality, King's College London
2018-19 **Teaching Assistant** Microeconomics 1, King's College London
2016 **Volunteer GCSE Maths Tutor** Action Tutoring
2015-16 **Masters Supervision** for MSc in Ecological Economics, University of Edinburgh.

INVITED TALKS & CONFERENCE PRESENTATIONS

- 2024 **International Conference** Royal Statistical Society
2023 **Medical Informatics Seminar** University of Edinburgh
2022 **National Strategic Network for Long COVID** NHS National Services Scotland
Academic GPs Meeting on Long COVID University of Edinburgh
Primary Care Practitioners' Session Public Health Scotland
2021 **Political Science & Political Economy Seminar** London School of Economics
2020 **Political Behaviour Colloquium** European University Institute
Global Online Meeting Economic Science Association
Early Career Network Conference Political Studies Association
Early Career Researcher Seminar University of Southampton
NORFACE TECHNO Group Norwegian University of Science and Technology
Quantitative Political Economy Seminar King's College London
Centre for Experimental Social Sciences Conference New York University
2019 **Centre for British Politics, PhD Conference** King's College London
2018 **AI Safety and Policy Event** Artificial Intelligence Strategy and Policy Network
2016 **Parliamentary Roundtable: How Should We Measure National Success?** UK Government
2015 **Five Headline Indicators of National Success: Report Launch** Royal Statistical Society
2014 **European Social Survey Policy Seminar** OECD

PROFESSIONAL MEMBERSHIPS

- 2024-present Fully Accredited Researcher under the Digital Economy Act (2017)
2021-present European Association of Labour Economists
2020-21 Royal Economic Society
American Economic Association
Economic Science Association
2011-14 Institute for Environmental Management and Assessment (Associate Member)
2010-11 International Society for Ecological Economics

PROFESSIONAL SERVICE

2021-present	Reviewer BMJ Open; Oxford Bulletin of Economics and Statistics; European Journal of Political Economy; European Economic Review; Political Science Research & Methods; Journal of IT & Politics.
2023-2024	Impact Committee Usher Institute, University of Edinburgh
2021-2023	Journal Club Organiser Medical Informatics Group, University of Edinburgh.
2018	Conference Organiser and Chair AI and Automation: Implications for researchers in the social sciences. One-day conference hosted at King's College London.
2018-20	Reading Group Organiser Formal Theory Group, King's College London.
2016	Event Organiser and Chair An evening in conversation with Ed Miliband MP and Nick Srnicek. The Future of Work: Harnessing new technologies to benefit all.
2014-16	Secretariat for the All-Party Parliamentary Group on Wellbeing Economics.

SELECTED TRAINING

2026	Generative AI for Data Science Mathematics & Computer Science, University of Glasgow
2024	Research, GDPR and Confidentiality (renewal) Medical Research Council
	Safe Researcher Training UK Data Service
	Data Science for Domain Scientists Data Science Unit, University of Edinburgh
2023	Aspiring Manager Program University of Edinburgh
2022	Machine Learning with Python ED-DaSH, University of Edinburgh
	High-Dimensional Statistics in R ED-DaSH, University of Edinburgh
	GIT for Me: Introduction to GitHub Jumping Rivers
2021	Managing Difficult Conversations NHS Professionals
2018	Introduction to R for Biostatistics and Health Informatics King's College London
	PhD Summer School: Applied Microeconomics University of Lancaster
2017	Principles of Economics with Calculus (MOOC) California Institute of Technology
2014	Introduction to Statistics Using SPSS Imperial College London
2013	Introduction to Computer Science and Programming in Python (MOOC) MIT

SOFTWARE SKILLS

R, L^AT_EX, Qualtrics

REFEREES

Available on request.